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1. $210 = 5 \times 2 \times 3 \times 7$

$$55 = 5 \times 11$$

$$\text{HCF} = 5$$

$$210 \times 5 + 55y = 5$$

$$1050 + 55y = 5$$

$$55y = 5 - 1050$$

$$y = \frac{-1045}{55}$$

$$y = -19$$

$$(1) -19$$

<sol_end>

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2. $a + b = 3 + 7 = 10$

$$10 = 2 \times 5$$

Least prime factor is 2.

$$(1) 2$$

<sol_end>

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3. Smallest two digit composites no. = 10

$$\text{Smallest composites no.} = 4$$

$$4 = 2 \times 2$$

$$10 = 2 \times 5$$

$$\text{LCM} = 2 \times 2 \times 5$$

$$= 20$$

$$(3) 20$$

<sol_end>

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4. $18 - 7 = 11$

$$21 - 10 = 11$$

$$24 - 13 = 11$$

$$18 = 2 \times 3 \times 3$$

$$21 = 3 \times 7$$

$$24 = 2 \times 2 \times 2 \times 3$$

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 7$$

$$= 8 \times 9 \times 7$$

$$= 504$$

$$504 - 11 = 493$$

$$(2) 493$$

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5. $p(y) = y^6 - 3y^4 + 2y^2 + 6$

$$y = 1$$

$$= 1^6 - 3(1)^4 + 2(1)^2 + 6$$

$$= 1 - 3 + 2 + 6$$

$$= -2 + 2 + 6$$

$$= 6$$

$$q(y) = y^5 - y^3 + 2y^2 + y - 6$$

$$y = 2$$

$$= (2)^5 - (2)^3 + 2(2)^2 + 2 - 6$$

$$= 32 - 8 + 8 + 2 - 6$$

$$= 42 - 14$$

$$= 28$$

$$p(1) = 6, \quad q(2) = 28$$

$$p(1) + q(2) = 6 + 28 = 34$$

$$(1) 34$$

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6. The zeroes are equal So,

$$b^2 - 4ac = 0$$

$$b = 3x,$$

$$a = 2$$

$$c = 8$$

$$\text{So, } (3x)^2 - 4(2)(8) = 0$$

$$9k^2 = 64$$

$$k^2 = \frac{64}{9}$$

$$k = \sqrt{\frac{64}{9}}$$

$$k = \pm \frac{8}{3}$$

$$(4) = \pm \frac{8}{3}$$

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7. We know, $aby = \frac{d}{a}$

$$\text{So, } (-1) By = \frac{c}{1}$$

$$By = -c \Rightarrow -By = C \quad \dots(1)$$

$$\text{and } x^3 + ax^2 + bx + c = 0$$

$$x = -1$$

$$(-1)^3 + a(-1)^2 + b(-1) + c = 0$$

$$-1 + a - b + c = 0$$

$$c = b - a + 1 \quad \dots(2)$$

equation (1) and (2),

$$-By = b - a + 1$$

$$By = -b + a - 1$$

$$By = a - b - 1$$

$$(4) a - b - 1$$

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8. $x^2 - 5x + 1 = 0$

Dividing both sides by x,

$$x - 5 + \frac{1}{x} = 0$$

$$x + \frac{1}{x} = 5$$

$$(1) 5$$

<sol_end>

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9. Given, $b^2 - 4ac = -68$

$$a = m$$

$$b = -4$$

$$c = 7$$

$$-68 = (-4)^2 - 4(m)(7)$$

$$-68 = 16 - 28m$$

$$-68 - 16 = -28m$$

$$\frac{-84}{-28} = m$$

$$m = 3$$

$$(4) 3$$

<sol_end>

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10. $(3) x^2 - 6x - 5 = 0$

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11. Let the integers be x and $x + 1$.

ATQ, $x^2 + (x + 1)^2 = 365$

$$x^2 + x^2 + 2x + 1 = 365$$

$$2x^2 + 2x + 1 - 365 = 0$$

$$2x^2 + 2x - 364 = 0$$

By 2,

$$x^2 + x - 182 = 0$$

$$x^2 + 14x - 13x - 182 = 0$$

$$x(x + 14) - 13(x + 14) = 0$$

$$(x - 13)(x + 14) = 0$$

$$\text{So, } x - 13 \text{ and } x = -14$$

It's a positive integer

So, the no. is 13 and the consecutive no. is 14.

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12. $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$

$$x^2 - px - p - x = 0$$

$$\text{So, } \alpha + \beta = \frac{-(-p)}{1} = p$$

$$\alpha\beta = \frac{-p-k}{1} = -p-k$$

$$(\alpha + 1)(\beta + 1) = 6$$

$$\alpha\beta + \alpha + \beta + 1 = 6$$

$$\alpha\beta + \alpha + \beta = 5$$

$$p - p - k = 5$$

$$-k = 5$$

$$k = -5$$

<sol_end>

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13. Let his present age be x,

$$\frac{1}{x-3} + \frac{1}{x+5} = \frac{1}{3}$$

$$\frac{x+5+x-3}{(x-3)(x+5)} = \frac{1}{3}$$

$$\frac{2x+2}{x^2-3x+5x-15} = \frac{1}{3}$$

$$3(2x+2) = x^2 + 2x - 15$$

$$6x + 6 - x^2 - 2x + 15 = 0$$

$$-x^2 + 4x + 21 = 0$$

$$x^2 - 4x - 21 = 0$$

$$x^2 - 7x + 3x - 21 = 0$$

$$x(x-7) + 3(x-7) = 0$$

$$\text{So, } x = -3 \text{ and } x = 7$$

Age can't be negative so, $x = 7$

$\therefore$ This present age is 7.

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$$14. \quad 40 = 2 \times 2 \times 2 \times 5$$

$$42 = 2 \times 3 \times 7$$

$$45 = 3 \times 3 \times 5$$

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7$$

$$= 8 \times 9 \times 35$$

$$= 72 \times 35$$

$$= 2520 \text{ cm}$$

Therefore each should walk 2520 cm So, that each can cover the same distance in complete steps.

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15. (i) By contradiction,

Let we assume is $\sqrt{5}$ rational

So, $\sqrt{5} = \frac{a}{b}$ (where a and b are co-primes and integers and $b \neq 0$)

$$\sqrt{5} b = a$$

Squaring both sides,

$$5b^2 = a^2 \quad \dots(1)$$

$$b^2 = \frac{a^2}{5}$$

By theorem:- If p divides a^2 it also divides a.

$$\Rightarrow \frac{a}{5}, 5 \text{ is a factor of } a \quad \dots(2)$$

Now let $\frac{a}{5} = c$ (c is any positive integer).

$$a = 5c$$

putting in (1),

$$5b^2 = (5c)^2$$

$$5b^2 = 25c^2$$

$$\frac{\cancel{5}b^2}{\cancel{25}} = c^2$$

$$\frac{b^2}{5} = c^2$$

By using the same above theorem,

$\Rightarrow \frac{b}{5}$, So, 5 is a factor of b ... (3)

From (2) and (3)

So, a and b are not co-prime no. 8 and

Hence our assumption is wrong and

$\sqrt{5}$ is an irrational no.

Hence proved

(ii) Let a be the positive integer and $b = 3$

We know $a = b_2 + r$,

$0 \leq r < b$

Now, $a = 3q + r$, $0 \leq r < b$

The possibilities of remainder is 0, 1 or 2

Case 1: when $a = 3q$

$$a^2 = (3q)^2 = 9q^2 = 3q \times 3q = 3m$$

where $m = 3q^2$

Case 2: when $a = 3q + 1$

$$\begin{aligned} a^2 &= (3q + 1)^2 \\ &= 9q^2 + 6q + 1 \\ &= 3q(3q + 2) + 1 \\ &= 3m + 1 \end{aligned}$$

Where $m = q(3q + 2)$

Case 3: when $a = 3q + 2$

$$\begin{aligned} a^2 &= (3q + 2)^2 \\ &= 9q^2 + 12q + 4 \\ &= 3q(3q + 4) + 1 \\ &= 3m + 1 \end{aligned}$$

Hence Proved

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16.

$$x^2 - 7x + 5x - 35 = 0$$

$$x(x - 7) + 5(x - 7) = 0$$

$$(x + 5)(x - 7) = 0$$

So,

$$x = -5 \text{ and } x = 7$$

Hence other zeroes are, - 5 and 7.

<sol_end>

<sol_start id=17>

17. (i) $60 = 2 \times 2 \times 3 \times 5$
 $84 = 2 \times 2 \times 3 \times 7$
 $108 = 2 \times 2 \times 3 \times 3 \times 3$
 $\text{HCF} = 2 \times 2 \times 3$
 $= 12$
 (2) 12

(ii) Hindi = $\frac{60}{12} = 5$ rooms

English = $\frac{84}{12} = 7$ rooms

Maths = $\frac{108}{12} = 9$ rooms

Total = $5 + 7 + 9$
 $= 21$ rooms

(4) 21

(iii) $60 = 2 \times 2 \times 3 \times 5$
 $84 = 2 \times 2 \times 3 \times 7$
 $108 = 2 \times 2 \times 3 \times 3 \times 3$
 $\text{LCM} = 2 \times 2 \times 3 \times 5 \times 7 \times 3$
 $= 60 \times 63$
 $= 3780$
 (1) 3780

(iv) $\text{HCF} \times \text{LCM} = 12 \times 3780$
 $= 45,360$
 (4) 45,360

(v) $108 = 2 \times 2 \times 3 \times 3 \times 3$

$$= 2^2 \times 3^3$$

$$(4) 2^2 \times 3^3$$

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<sol_start id=18>

18. (i) (1) -2, 8

$$(ii) (2) -x^2 + 6x + 16$$

$$(iii) = -x^2 + 6x + 16$$

$$x = 4$$

$$= -(4)^2 + 6(4) + 16$$

$$= -16 + 24 + 16$$

$$= 24$$

$$(3) 24$$

$$(iv) = -x^2 + 3x - 2$$

$$= x^2 - 3x + 2$$

$$= x^2 - 2x - x + 2$$

$$= x(x - 2) - 1(x - 2)$$

$$= (x - 1)(x - 2)$$

$$\text{So, } x = 1 \text{ and } 2$$

$$(4) 1, 2$$

$$(v) \alpha + \beta = 4 - 3 = 1$$

$$\alpha\beta = -12$$

$$\text{So, } x^2 - x + 24$$

$$(1) x^2 - x + 24$$

<sol_end>

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